

The cooked meat-derived mammary carcinogen 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine promotes invasive behaviour of breast cancer cells.

The cooked meat derived genotoxic carcinogen 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) induces cancer of the colon, prostate and mammary gland when fed to rats. Epidemiology studies link these tumours to a Western diet and exposure to heterocyclic amines such as PhIP. We have shown that PhIP is also potently estrogenic and have proposed that this hormonal activity contributes to its target site carcinogenicity. We now postulate that the estrogenic properties of PhIP influence metastatic potential. We have used an in vitro assay for cell invasion based upon digestion and migration through a reconstituted basement membrane model. Zymography and immunoblotting were used to confirm PhIP-mediated changes associated with induction of the invasive phenotype. Treatment of the mammary cancer cell lines MCF-7 and T47D with PhIP induces cells to digest and migrate through a reconstituted basement membrane. The response was dose dependent, observed at sub-nanomolar concentrations of PhIP and was inhibited by the antiestrogen ICI 182,780. The PhIP-induced invasive phenotype was associated with expression of cathepsin D, cyclooxygenase-2 and matrix metalloproteinase activity. These findings emphasise the range and potency of the biological activities associated with this cooked meat product and mechanistically support the tissue-specific carcinogenicity of the chemical. Copyright © 2010 Elsevier Ireland Ltd. All rights reserved.